Statistical image reconstruction:
An analysis of the trade-off between bias and variance in PET-scans

J. Lombaerts, Promotor: Prof. Dr. M. Defrise

Department of Nuclear Medicine, Vrije Universiteit Brussel

PET (Positron Emission Tomography) scans are used in medicine to visualise a patient's interior and rely on the indirect emission of photons which can be observed using dedicated detectors.
Different analytical and statistical methods exist to reconstruct an image from the data. Historically the first reconstruction methods to be used were analytical methods such as filtered backprojection but as capable computers have become commercially available analytical methods have been deprecated in favour of statistical methods implementing iterative algorithms.
A Maximum Likelihood - Expectation Maximisation (MLEM) iterative image reconstruction algorithm was first proposed by Shepp and Vardi [1]. MLEM assumes a Poisson data distribution and remains to this day the default PET reconstruction algorithm. An alternative reconstruction algorithm was derived by M.E. Daube-Witherspoon and G. Muehllehner [2], the Image Space Reconstruction Algorithm (ISRA), which assumes a Gaussian data distribution. ISRA is seldom used and has not been studied as much as MLEM.
The goal was to study ISRA and to compare ISRA and MLEM both quantitatively (expected value, bias and variance) and qualitatively (do the images reconstructed with ISRA resemble the images reconstructed with MLEM).

The studies and comparisons were performed by reconstructing simulated data representing a small 2 pixel toy problem (fig. 1) and a realistic 512 pixel x 512 pixel Jaszczak phantom image (fig. 2). Both images were reconstructed using MLEM and ISRA. The differences in reconstruction quality were minimal, but noticeable. In particular ISRA suffered from a slightly decreased reconstruction quality on the outer edge of the image. These statistically noticeable differences were barely noticeable when visually examining the reconstructed images. Based on the limited sample of reconstructed images it can be concluded that for all practical purposes, MLEM and ISRA can be considered to achieve comparable reconstruction qualities, though further work is needed to strengthen this claim.

Figure 1: *A 2 pixel toy problem on which different image reconstruction methods were tested.*

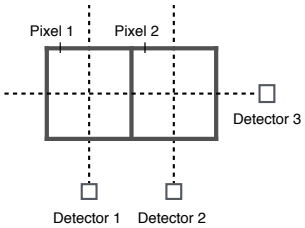
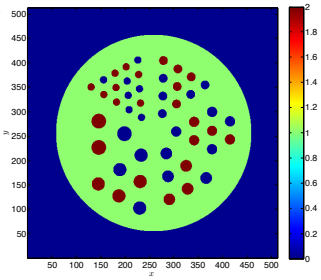


Figure 2: *The true image of a Jaszczak phantom which was reconstructed with MLEM and ISRA.*



References
[1] Shepp L.A., Vardi Y., Maximum likelihood estimation for emission tomography , (1982), IEEE Trans Med Imaging 1: 113 - 121
[2] Daube-Witherspoon M.E., Muehllehner G., An Iterative Image Space Reconstruction Algorithm Suitable for Volume ECT, (1986), IEEE Trans Med Imaging vol. mi-5, no. 2: 61 - 66